CSE251: Electronic Devices and Circuits

A. Course General Information:

Course Code:	CSE251
Course Title:	Electronic Devices and Circuits
Credit Hours (Theory + Lab):	3+0
Contact Hours (Theory + Lab):	3+3
Category:	Lecture/laboratory
Type:	Core
Prerequisite:	CSE250

B. Course Catalog Description (Content):

This course provides a comprehensive introduction to semiconductor devices, non-linear electronic circuits, and linear integrated circuits, combining theoretical mathematical modeling with fundamental circuit analysis techniques. The curriculum opens with an overview of solid-state device history and basic circuit representation using Kirchhoff's Current Law (KCL), Kirchhoff's Voltage Law (KVL), and Nodal Analysis. It then proceeds into an in-depth study of Operational Amplifiers (Op-Amps), covering open-loop and closed-loop configurations, Voltage Transfer Characteristics (VTC), negative feedback, inverting and non-inverting amplifiers, voltage followers, weighted summers, subtractors, differentiators, integrators, and non-linear circuits like logarithmic and exponential converters. The course then transitions to non-linear semiconductor physics and I-V characteristics, focusing on the Shockley diode equation, diode logic gates, ideal diode models, and Constant Voltage Drop (CVD and CVD+r) modeling, alongside circuit analysis using the Method of Assumed States. Applications in power conversion, such as half-wave and full-wave rectifiers, smoothing capacitors, peak-to-peak ripple, and ripple factor calculations, are thoroughly evaluated. Finally, the focus shifts to solid-state switches and amplifiers, examining Bipolar Junction Transistors (BJTs) and Field-Effect Transistors (MOSFETs, including n-channel and p-channel enhancement and depletion types). Topics include physical device operation across cut-off, active or triode, and saturation modes, piecewise linear modeling, switch and S-model representations, logic gate synthesis, transistor circuit configurations, and static DC circuit analysis using the Method of Assumed States.

C. Course Objective

a. Introduce the Operational Amplifier, its application as a comparator, and its implementation in linear and non-linear analog signal-processing circuits (such as summing, subtracting, integrating, and differentiating circuits).
b. Introduce fundamental electronic devices (diodes and transistors) along with the semiconductor physics principles governing their behavior.
c. Apply Piecewise-Linear modeling, ideal models, and the Method of Assumed States to analyze circuits containing non-linear devices.
d. Demonstrate the application of semiconductor diodes in designing logic gates, rectifiers, and voltage regulators.
e. Analyze the switching and amplification behavior of BJT and MOSFET circuits using DC biasing, load lines, and equivalent circuit models.
f. Train students to construct, analyze, and prototype non-linear and transistor-level electronic circuits in a weekly laboratory setting.
g. Develop students' ability to measure, test, and troubleshoot electronic circuit performance using standard laboratory instruments.
h. Enhance teamwork and technical communication skills through collaborative laboratory experiments and formal report writing.

D. Course outcome

Upon successful completion of this course, students will be able to

Sl.	CO Description	Weightage (%)
CO1	Explain and compare the underlying physical characteristics and operational principles of electronic devices, including Op-Amps, semiconductor diodes, BJTs, and MOSFETs.	10
CO2	Analyze the behavior of electronic circuits consisting of different non-linear electronic devices such as Diodes, OP-AMPs, BJTs, and MOSFETs using piecewise linear modeling, the method of assumed states, and general circuit analysis techniques.	35
CO3	Design and synthesize analog signal-processing and power-conversion circuits, such as amplifiers, comparators, rectifiers, voltage regulators, and transistor switches, to meet specified criteria.	30
CO4	Function effectively as a member of a team to perform collaborative	6

	non-linear electronic circuit experiments and hardware laboratory setups.	
CO5	Document and communicate experimental results, data analysis, and engineering insights clearly by preparing structured technical lab reports.	5
CO6	Apply modern electronic instruments and laboratory techniques to build, test, measure, and troubleshoot non-linear diode and transistor circuits individually under timed examination conditions.	14

E. Mapping of CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool:

Sl.	**CO Description**	**POs**	**Bloom's taxonomy domain/level**	**Delivery methods and activities**	**Assessment tools**	**KPIs of relevant PO**	**WK (WK1-WK9)**	**CEP(P1-P7)**	**SDG**	**CEA (A1-A5)**
CO1	Explain and compare the underlying physical characteristics and operational principles of electronic devices, including Op-Amps, semiconductor diodes, BJTs, and MOSFETs.	PO1	Cognitive / C2 (Understand)	Lectures, Interactive Discussions, Multimedia Slides	Quizzes, Midterm Exam, Final Exam	PO1(c): Apply knowledge of engineering fundamentals	WK3	N/A	N/A	N/A
CO2	Analyze the behavior of electronic circuits consisting of different non-linear electronic devices such as Diodes, OP-AMPs, BJT, and MOSFETs using piece-wise linear modeling, the method of assumed states, and general circuit analysis techniques.	PO2	Cognitive / C4 (Analyze)	Lectures, Problem Solving Sessions, Whiteboard Derivations	Quizzes, Midterm Exam, Final Exam	PO2(d): Analyze complex engineering problems using first principles	WK4	WP1, WP3	N/A	N/A
CO3	Design and synthesize analog signal-processing and power-conversion circuits, such as	PO3	Cognitive / C6 (Create)	Design Lectures, Guided Circuit Synthesis	Quizzes, Midterm Exam, Final Exam, Design	PO3(b): Design systems, components, or processes	WK5	WP1, WP2, WP3	SDG 7 (Affordable and Clean Energy), SDG 9	N/A

	amplifiers, comparators, rectifiers, voltage regulators, and transistor switches, to meet specified criteria.				Problems	to meet identified need			(Industry, Innovation, Infrastructure)	
CO4	Function effectively as a member of a team to perform collaborative non-linear electronic circuit experiments.	PO8	Affective / A3 (Valuing)	Hands-on Group Experiments, Collaborative Lab Work	Weekly In-lab Performance, Hardware Execution	PO8(b): Function effectively as a member in diverse teams	WK9	N/A	N/A	N/A
CO5	Document and communicate experimental results, data analysis, and engineering insights clearly by preparing structured technical lab reports.	PO9	Affective / A4 (Organization)	Lab Report Guidelines, Feedback on Drafts	Lab Reports	PO9(a): Communicate effectively through writing effective reports	WK9	N/A	N/A	N/A
CO6	Apply modern electronic instruments and laboratory techniques to build, test, measure, and troubleshoot electronic circuits individually.	PO5	Psychomotor / P4 (Complex Overt Response)	Equipment Demonstration, Open Lab Practice	Lab Test 1, Lab Test 2	PO5(c): Apply modern engineering and IT tools	WK6	N/A	N/A	EA1, EA5

F. Course Materials:

i. Text and Reference Books:

Sl.	Title	Author(s)	Publication Year	Edition	Publisher	ISBN
1	Foundations of Analog and Digital Electronic Circuits	Anant Agarwal, Jeffrey H. Lang	2005	1st ed.	Morgan Kaufmann Publishers	978–1–55–860735–4

2	Microelectronic Circuits	Adel S. Sedra, Kenneth C. Smith	2015	7th ed.	Oxford University Press	978–0–19–933913–6
3	The Art of Electronics	Paul Horowitz, Winfield Hill	2016	3rd ed.	Cambridge University Press	978–0–521–80926–9
4	Operational Amplifiers and Linear Integrated Circuits	Robert F. Coughlin, Frederick F. Driscoll	2001	6th ed.	Prentice Hall	978–0–130–14991–6

ii. Other materials (if any)

1. Lecture Slides
2. Lecture Notes/Handouts
3. Video Lectures
4. Practice Sheets
5. Previous Semesters' Mid and Final Questions
6. Lab Sheets
7. Simulation Tool (LTspice)

G. Lesson Plan:

No	Topic	Week/Lecture#	Related CO (if any)
1	History and importance of electronic devices, transition from mechanical switches to vacuum tubes to solid-state devices, current state of the art in semiconductor technology	Lecture 1	CO1
2	Alternative circuit representation, review of Kirchhoff's Current Law (KCL), Kirchhoff's Voltage Law (KVL), and Nodal Analysis	Lecture 2	CO1, CO2
3	Introduction to Operational Amplifiers, differential amplifiers, solving Op-Amp-based circuits using KCL, KVL, and Nodal Analysis, Op-Amp circuits in open-loop configuration, characteristics of infinite gain, Voltage Transfer Characteristics (VTC)	Lecture 3	CO1, CO2

4	Op-Amp circuits in closed-loop configuration, controlling gain through negative feedback, virtual ground concept, inverting amplifier, non-inverting amplifier, voltage follower, weighted summer, subtractor, differentiator, integrator, VTC graphs of inverting and non-inverting amplifiers	Lecture 4	CO2, CO3
5	Advanced Op-Amp applications, exponential converters, logarithmic converters, multipliers, and dividers	Lecture 5	CO2, CO3
-	Quiz 1 (Lectures 2 to 5)	-	-
6	Introduction to I-V characteristics of linear elements (resistors, voltage sources, current sources) and degenerate elements (open circuit, short circuit), introduction to non-linear devices, physical operation of semiconductor diodes, Shockley diode equation, diode logic gates	Lecture 6	CO1
7	Modeling semiconductor diodes, ideal diode model, Constant Voltage Drop (CVD) model, CVD with series resistance (CVD+r) model, introduction to the Method of Assumed States	Lecture 7	CO1, CO2
8	Solving complex non-linear diode circuits using the Method of Assumed States, problem-solving session	Lecture 8	CO2
9	Diode applications, diode logic realization, and introduction to half-wave and full-wave rectifiers.	Lecture 9	CO2, CO3
-	Quiz 2 (Lectures 6 to 9)	-	-
10	Pre-midterm review, comprehensive problem solving on Op-Amps, diode models, and the Method of Assumed States	Lecture 10	CO1, CO2, CO3
-	Midterm Exam (Lectures 1 to 10)	-	-
11	Introduction to rectifiers, half-wave rectifiers and transfer characteristics, full-wave rectifier (bridge configuration) and transfer characteristics	Lecture 11	CO1, CO3
12	Rectifiers revisited, average value of output voltage, smoothing capacitor filters, peak-to-peak ripple voltage, ripple factor calculations	Lecture 12	CO2, CO3

13	Zener diode physical characteristics, reverse breakdown region, Zener diode modeling, voltage regulation using Zener diodes, line and load regulation analysis	Lecture 13	CO1, CO2, CO3
14	Introduction to solid-state switches, basic inverter concepts, controlled current sources, introduction to Bipolar Junction Transistors (BJTs), physical operation of NPN and PNP transistors	Lecture 14	CO1
15	BJT operational modes (cut-off, active, saturation), switch model (S-model) of BJT, designing logic gates with BJTs, ideal output characteristics	Lecture 15	CO1, CO3
16	Piecewise linear modeling of BJTs, non-ideal static analysis, BJT circuit configurations (Common Emitter), DC biasing, load line analysis, solving BJT circuits using the Method of Assumed States	Lecture 16	CO2, CO3
-	Quiz 3 (Lectures 11 to 16)	-	-
17	Introduction to Field-Effect Transistors (FETs), physical structure and operation of MOSFETs, n-channel and p-channel enhancement-type and depletion-type MOSFETs	Lecture 17	CO1
18	Operational regions of MOSFETs (cut-off, triode, saturation), S-model of MOSFET as a switch, designing logic gates with MOSFETs, ideal output characteristics	Lecture 18	CO1, CO3
19	Piecewise linear modeling of MOSFETs, non-ideal static analysis, MOSFET circuit configurations, DC biasing methods	Lecture 19	CO2
20	Solving MOSFET circuits using the Method of Assumed States, comprehensive transistor problem solving	Lecture 20	CO2, CO3
-	Quiz 4 (Lectures 17 to 20)	-	-
21	Comprehensive course review, final exam prep, synthesis of Op-Amp, diode, BJT, and MOSFET circuits	Lecture 21	CO1, CO2, CO3
-	Final Exam (Lectures 11 to 20)	-	-

H. Lab plan

Lab	Topics Covered	Lab Activities	Submission / Evaluation
Lab 1 *(Week 2)*	Operational Amplifier Applications & Characteristics	Hardware assembly of op-amp comparator, non-inverting amplifier, inverting amplifier, and summing amplifier; measurement and plotting of voltage transfer characteristics (VTC).	In-lab Performance & Lab Report 1
Lab 2 *(Week 3)*	Semiconductor Diodes & Logic Circuits	Measurement of I-V characteristics of a diode; construction and testing of diode-based logic gates and clipping/clamping circuits.	In-lab Performance & Lab Report 2
Lab 3 *(Week 4)*	Pre-Assessment Practice	Open practice session for reviewing prior hardware experiments and troubleshooting circuit builds.	Practice & Self-Assessment
Lab 4 *(Week 5)*	Midterm Practical Assessment	Individual hardware implementation of assigned circuit problems and experimental verification under exam conditions.	Lab Test 1
Lab 5 *(Week 7)*	Diode Rectifier Circuits (Half-Wave & Full-Wave)	Hardware setup of half-wave and full-wave rectifiers; measurement of ripple factor, rectification efficiency, and output waveforms using an oscilloscope.	In-lab Performance & Lab Report 3
Lab 6 *(Week 8)*	BJT I-V Characteristics & Logic Circuits	Measurement of BJT input/output characteristics across operating regions (cutoff, active, saturation); implementation of BJT switch and logic functions.	In-lab Performance & Lab Report 4

Lab 7 *(Week 9)*	MOSFET I-V Characteristics & Logic Functions	Experimental extraction of MOSFET $I_D - V_{DS}$ and $I_D - V_{GS}$ transfer curves; hardware implementation of basic logic gates using NMOS/PMOS configurations.	In-lab Performance & Lab Report 5
Lab 8 *(Week 10)*	Final Assessment Practice	Open practice session for reviewing transistor/rectifier circuit setups, signal measurement, and hardware troubleshooting.	Practice & Self-Assessment
Lab 9 *(Week 11)*	Final Practical Assessment	Individual hardware build, circuit debugging, and performance evaluation under exam conditions.	Lab Test 2

Lab CO information:

COs	Category	Criteria	Total Marks
CO4	Continuous Assessment	Weekly In-lab Group Collaboration, Work Distribution, and Team Performance	6%
CO5	Continuous Assessment	Weekly Lab Reports, Data Analysis, Graphical Plotting, and Technical Documentation	5%
CO6	Summative Assessment	Practical Examinations (Lab Test 1 & Lab Test 2): Individual Circuit Assembly, Measurement, and Troubleshooting	14%

I. Assessment Tools:

Assessment Tools	Weightage (%)
1. Attendance	5
2. Assignment	5

3. Quizzes	15
4. Mid Term Examination	25
5. Lab	25
6. Final	25

J. CO Assessment Plan:

Assessment Tools						
	CO1	**CO2**	**CO3**	**CO4**	**CO5**	**CO6**
Quizzes	√	√	√			
Assignments	√	√	√			
Midterm Exam	√	√	√			
Final Exam	√	√	√			
In-lab Work / Performance				√		
Lab Report					√	
Lab Tests (1 & 2)						√

K. CO Attainment Policy:

As per the Department of CSE course outcome attainment policy

L. Grading policy:

As per the BRAC University grading policy

M. Course Coordinator:

Purbayan Das, Moynul Asik Moni